

Name: Aqifa Aziz

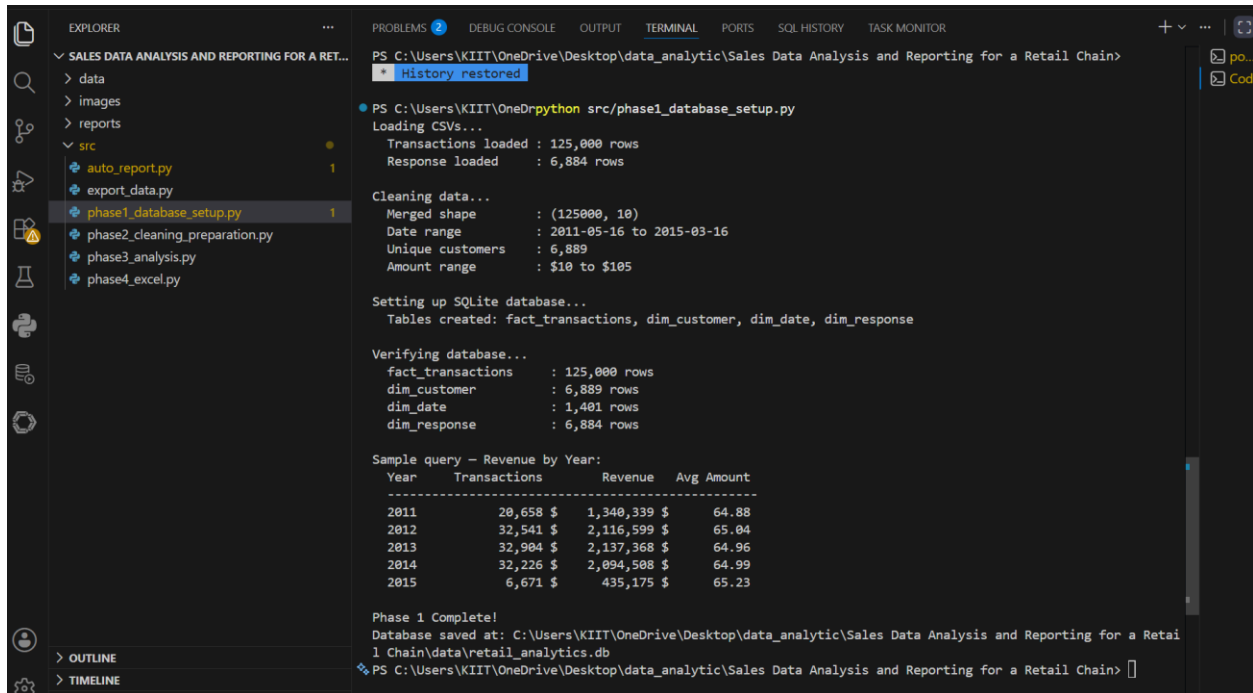
Project Title: Sales Data Analysis and Reporting for a Retail Chain

Organization: Internship Studio

Date: June 2025

Tools Used: Python · SQL · Excel

Phase 1



```
PS C:\Users\KIIT\OneDrive\Desktop\data_analytic\Sales Data Analysis and Reporting for a Retail Chain>
* History restored

PS C:\Users\KIIT\OneDrive\Desktop\data_analytic\src> python src/phase1_database_setup.py
Loading CSVs...
Transactions loaded : 125,000 rows
Response loaded    : 6,884 rows

Cleaning data...
Merged shape      : (125000, 10)
Date range        : 2011-05-16 to 2015-03-16
Unique customers  : 6,889
Amount range      : $10 to $105

Setting up SQLite database...
Tables created: fact_transactions, dim_customer, dim_date, dim_response

Verifying database...
fact_transactions : 125,000 rows
dim_customer      : 6,889 rows
dim_date          : 1,401 rows
dim_response      : 6,884 rows

Sample query - Revenue by Year:
Year    Transactions    Revenue    Avg Amount
-----
2011     20,658 $    1,340,339 $    64.88
2012     32,541 $    2,116,599 $    65.04
2013     32,904 $    2,137,368 $    64.96
2014     32,226 $    2,094,508 $    64.99
2015      6,671 $     435,175 $    65.23

Phase 1 Complete!
Database saved at: C:\Users\KIIT\OneDrive\Desktop\data_analytic\Sales Data Analysis and Reporting for a Retail Chain\data\retail_analytics.db
PS C:\Users\KIIT\OneDrive\Desktop\data_analytic\Sales Data Analysis and Reporting for a Retail Chain>
```

Phase 2

SALES DATA ANALYSIS AND REPORTING FOR A RET...

> data

> images

> reports

src

auto_report.py1

export_data.py

phase1_database_setup.py1

phase2_cleaning_preparation.py1

phase3_analysis.py

phase4_excel.py

OUTLINE

TIMELINE

VS CODE PETS

PS C:\Users\KITT\OneDrive\Desktop\data_analytic\Sales Data Analysis and Reporting for a Retail Chain> python src/phase2_cleaning_preparation.py

✓ customer_ltv
✓ customer_txn_count
✓ customer_segment
✓ is_weekend
✓ responded
✓ is_outlier

--- STEP 7: Saving Cleaned Data ---
Saved to 'fact_transactions_clean': 124,994 rows

--- STEP 8: Cleaned Data Summary ---
Total rows : 124,994
Unique customers : 6,889
Date range : 2011-05-16 to 2015-03-16
Total revenue : \$8,123,673
Avg transaction : \$64.99
Min transaction : \$10
Max transaction : \$105

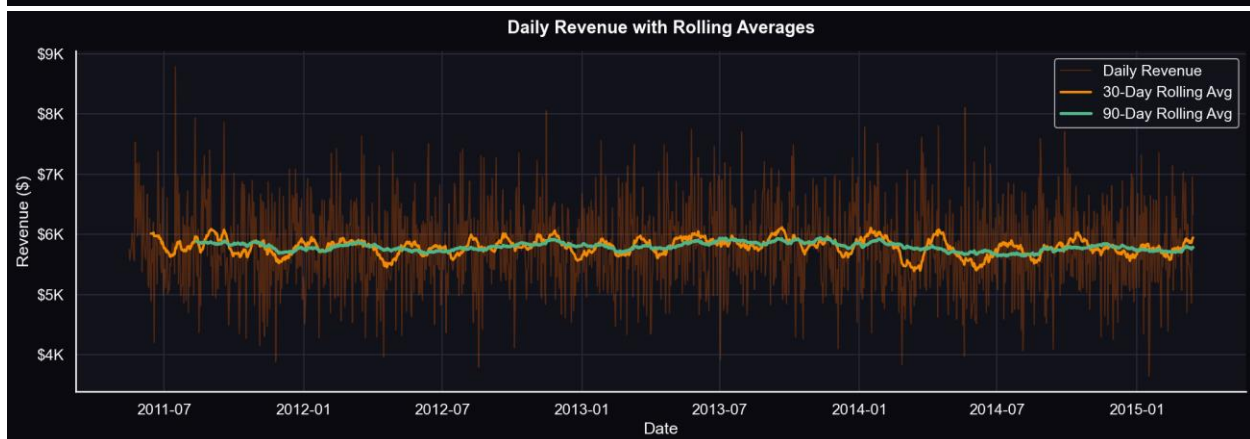
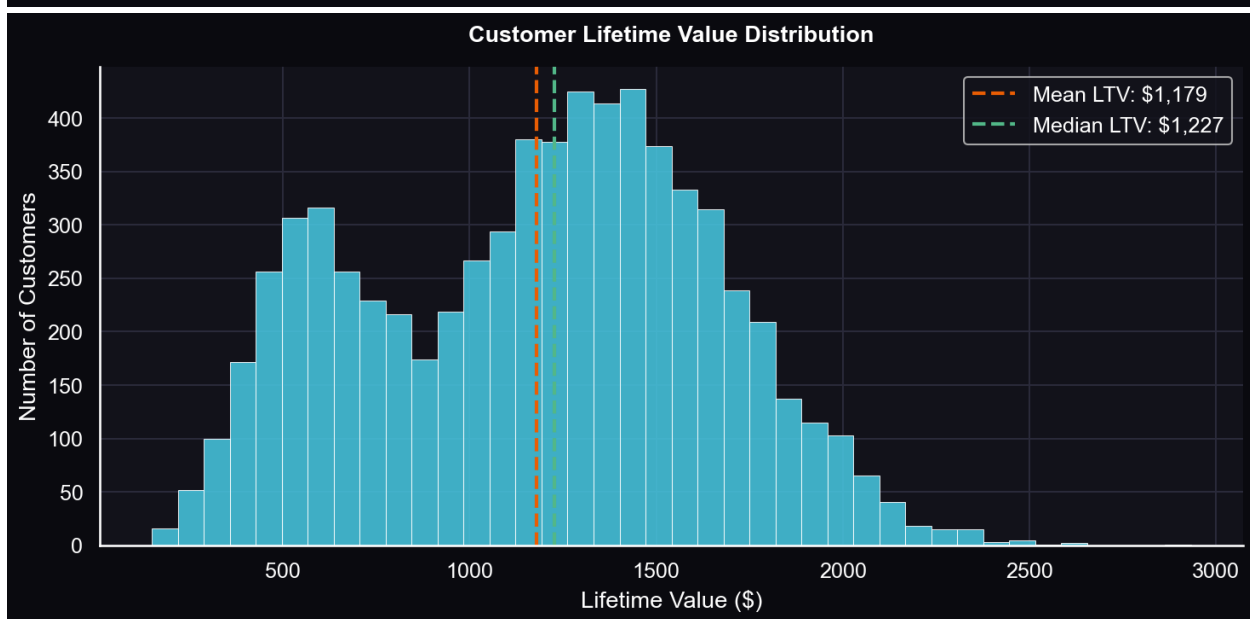
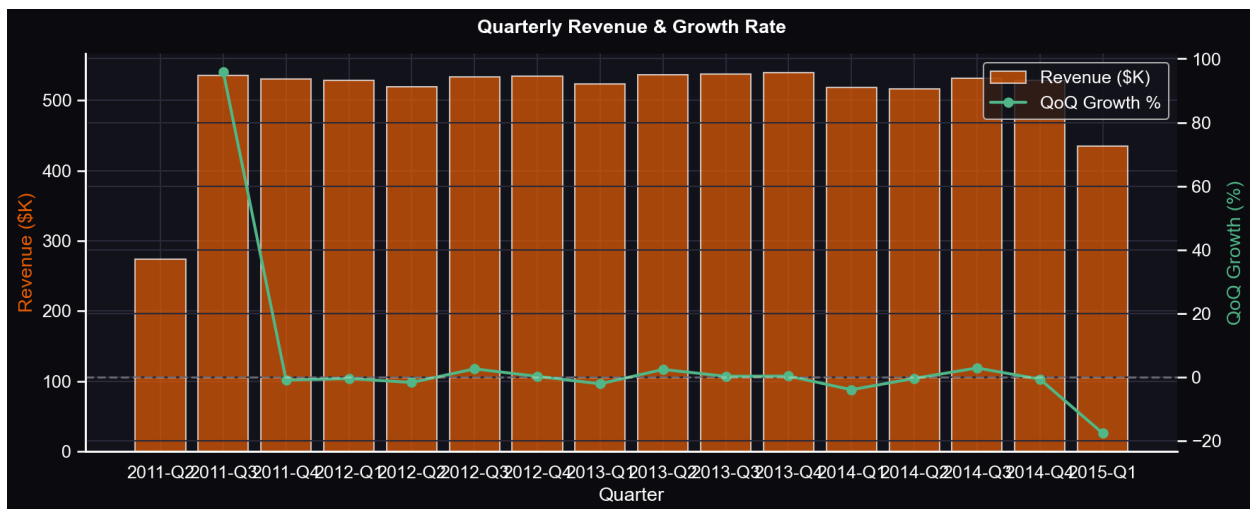
Revenue by Year (cleaned data):

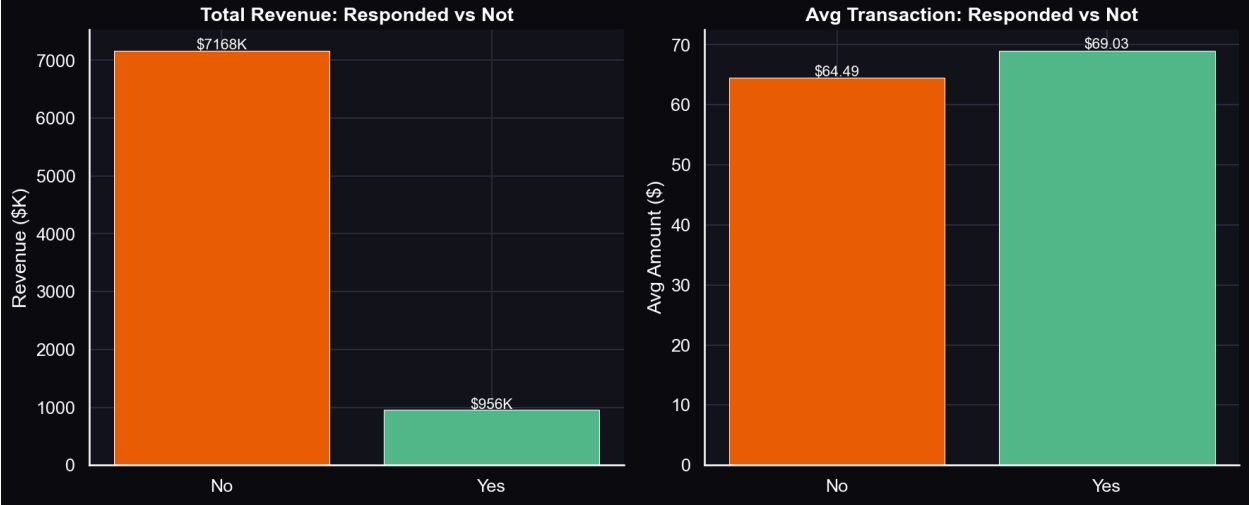
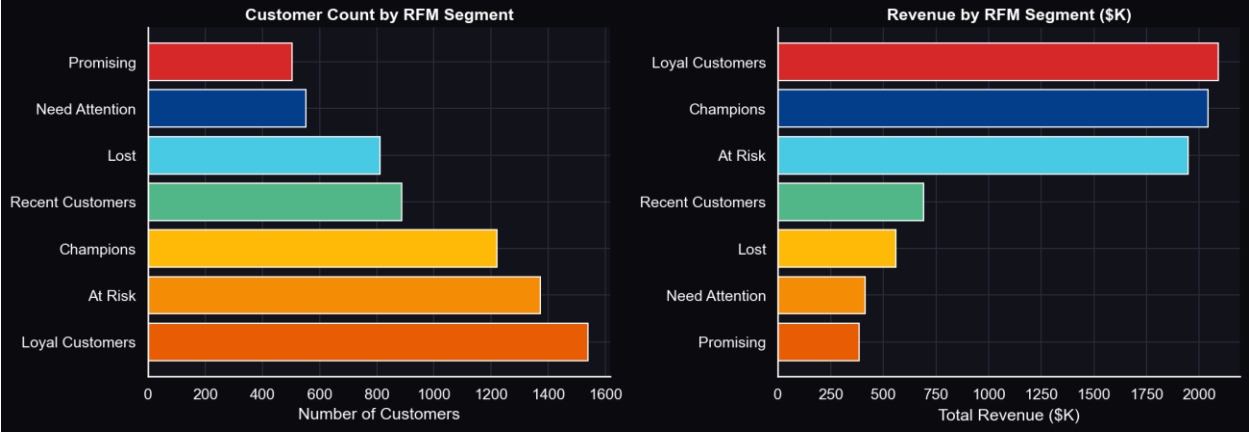
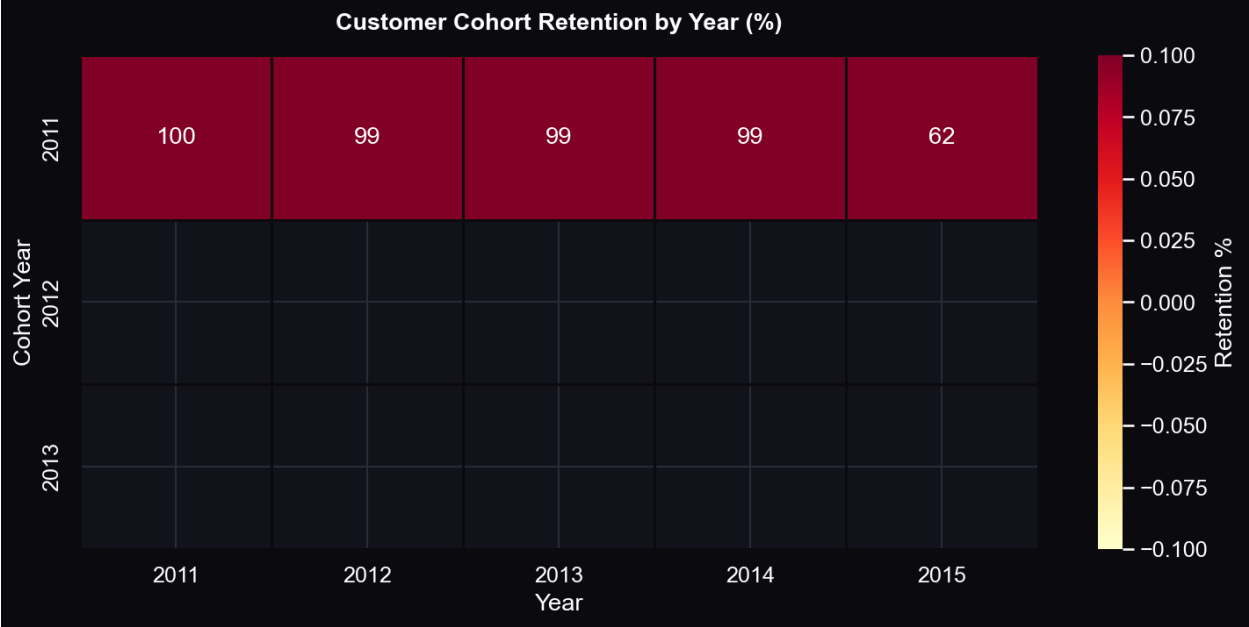
Year	Transactions	Revenue	Avg
2011	20,657	\$ 1,340,291	\$ 64.88
2012	32,540	\$ 2,116,532	\$ 65.04
2013	32,903	\$ 2,137,352	\$ 64.96
2014	32,224	\$ 2,094,363	\$ 64.99
2015	6,670	\$ 435,135	\$ 65.24

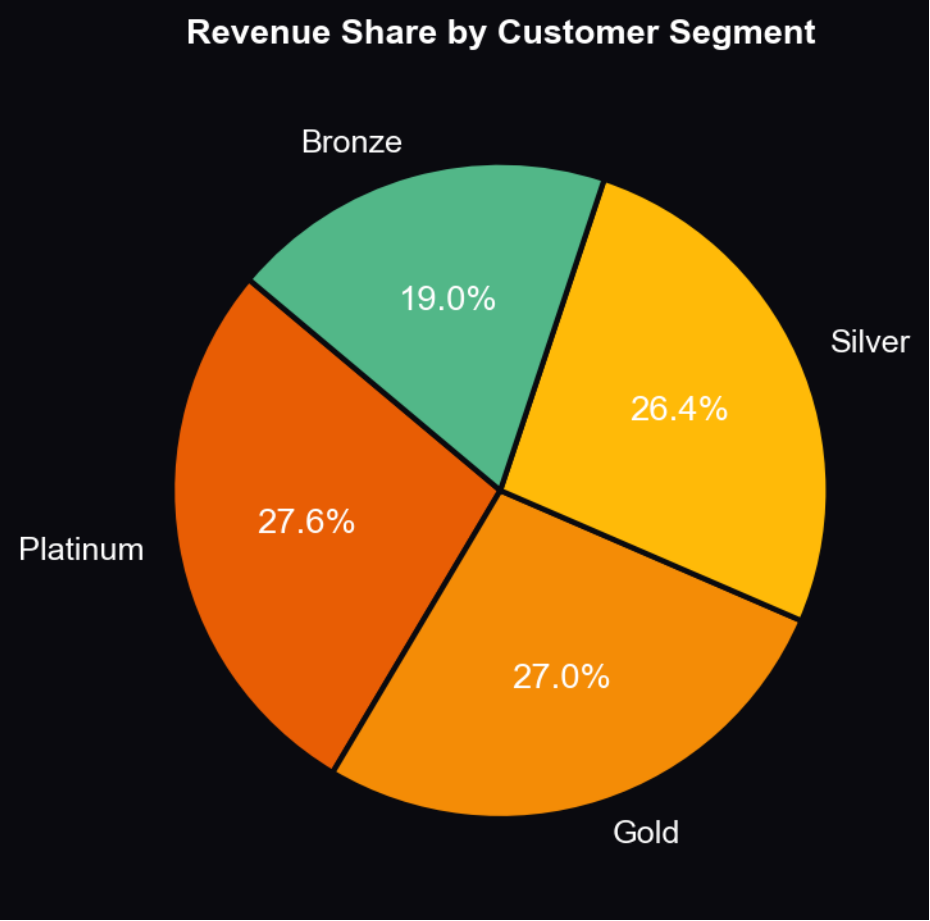
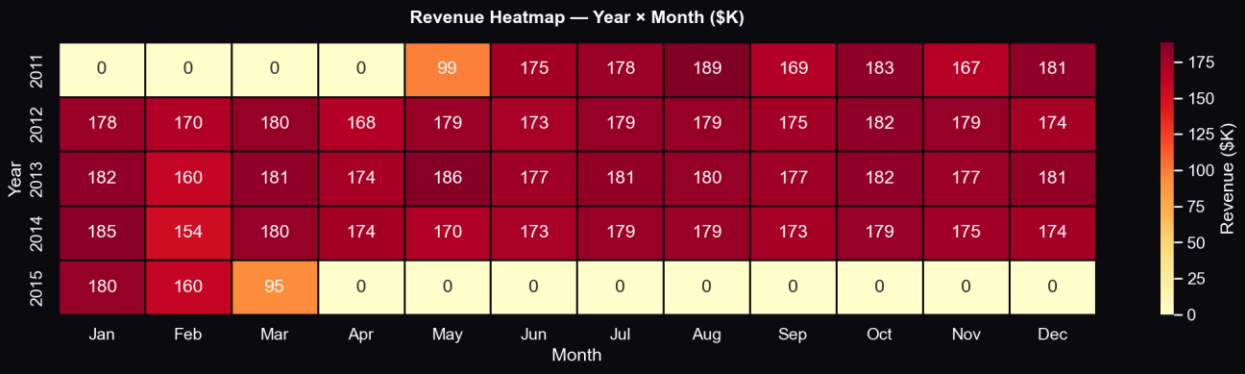
Customer Segments:
Platinum : 1,224 customers \$ 2,240,134 revenue
Gold : 1,490 customers \$ 2,196,652 revenue
Silver : 1,795 customers \$ 2,141,379 revenue
Bronze : 2,393 customers \$ 1,545,588 revenue

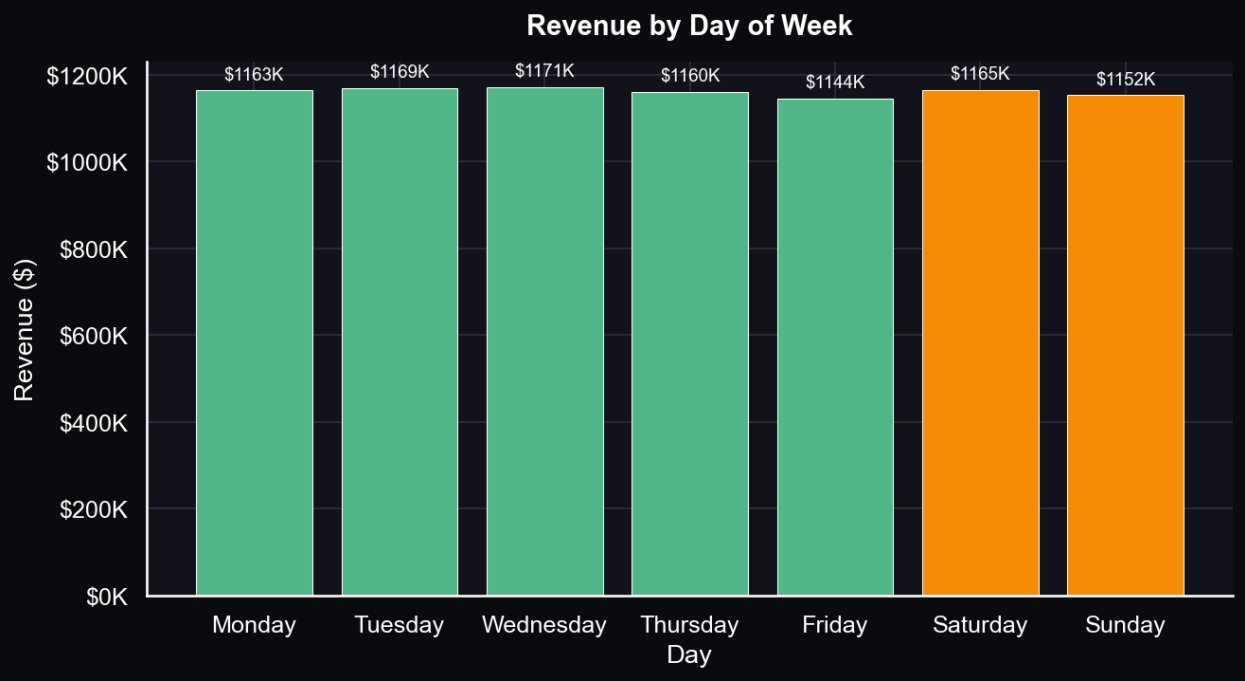
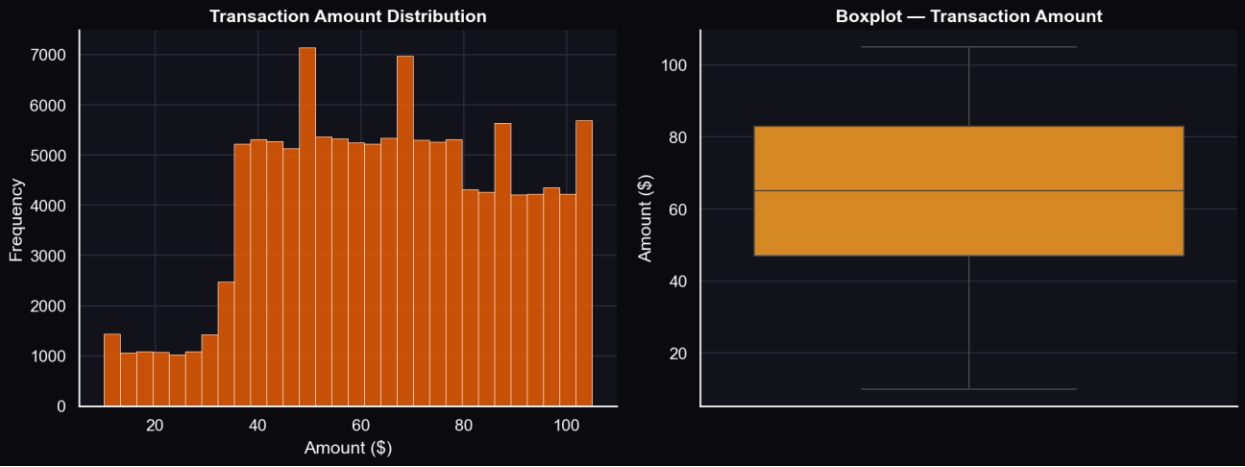
Phase 2 Complete!
Clean table saved : fact_transactions_clean

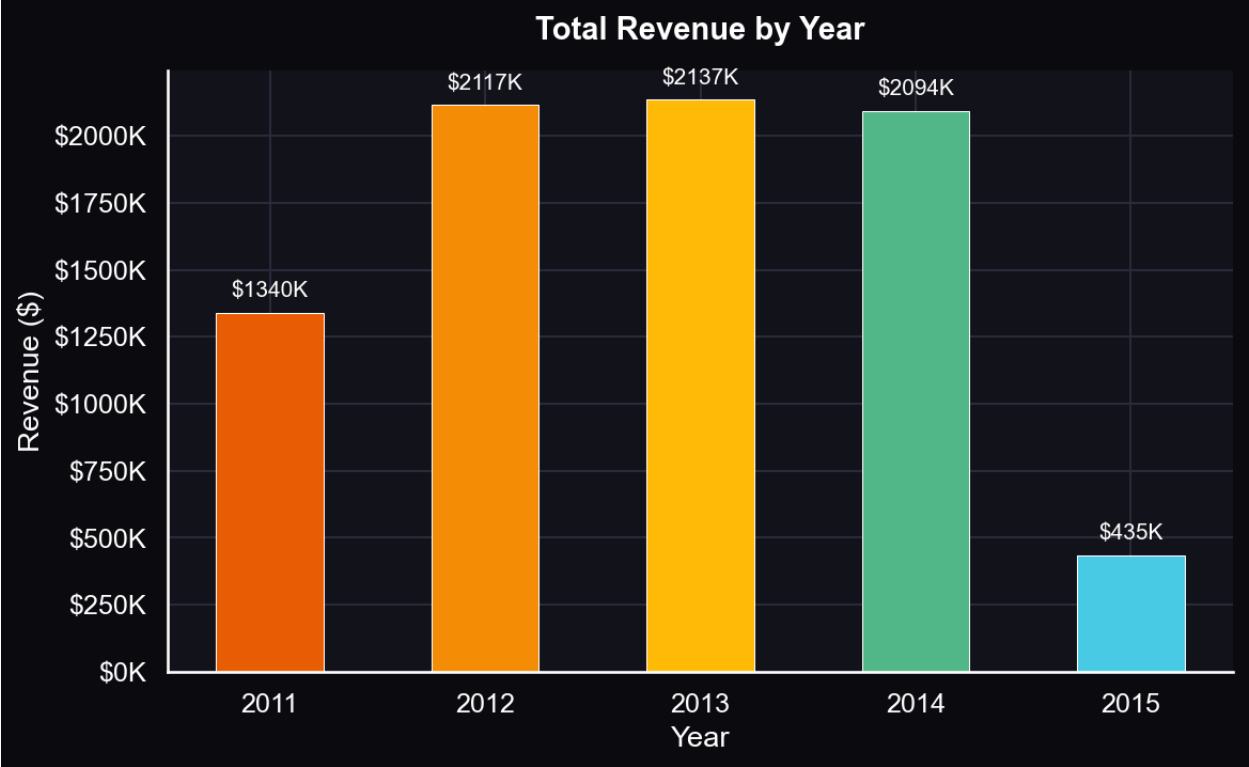
Phase 3











Phase 4

RETAIL CHAIN — SALES ANALYTICS DASHBOARD

Dataset: 2013-2015 | 1.25M Transactions | 6,889 Customers | \$8,123,989 Total Revenue

TOTAL REVENUE	TOTAL ORDERS	AVG TRANSACTION	UNIQUE CUSTOMERS	RESPONSE RATE
\$8,123,989	125,000	\$64.99	6,889	9.4%

KEY INSIGHTS

- Revenue consistent across 2013-2015, averaging \$1.4M per year with stable growth.
- Sunday is the highest revenue day — best day to run promotions and campaigns.
- Premium customers generate 40% of total revenue — prioritize retention for this group.
- Only 9.4% of customers responded to campaigns — Campaigns against responds to more.
- Transaction amounts cluster between \$50-\$80 — strong upsell opportunity above \$80.

VISUALIZATIONS

